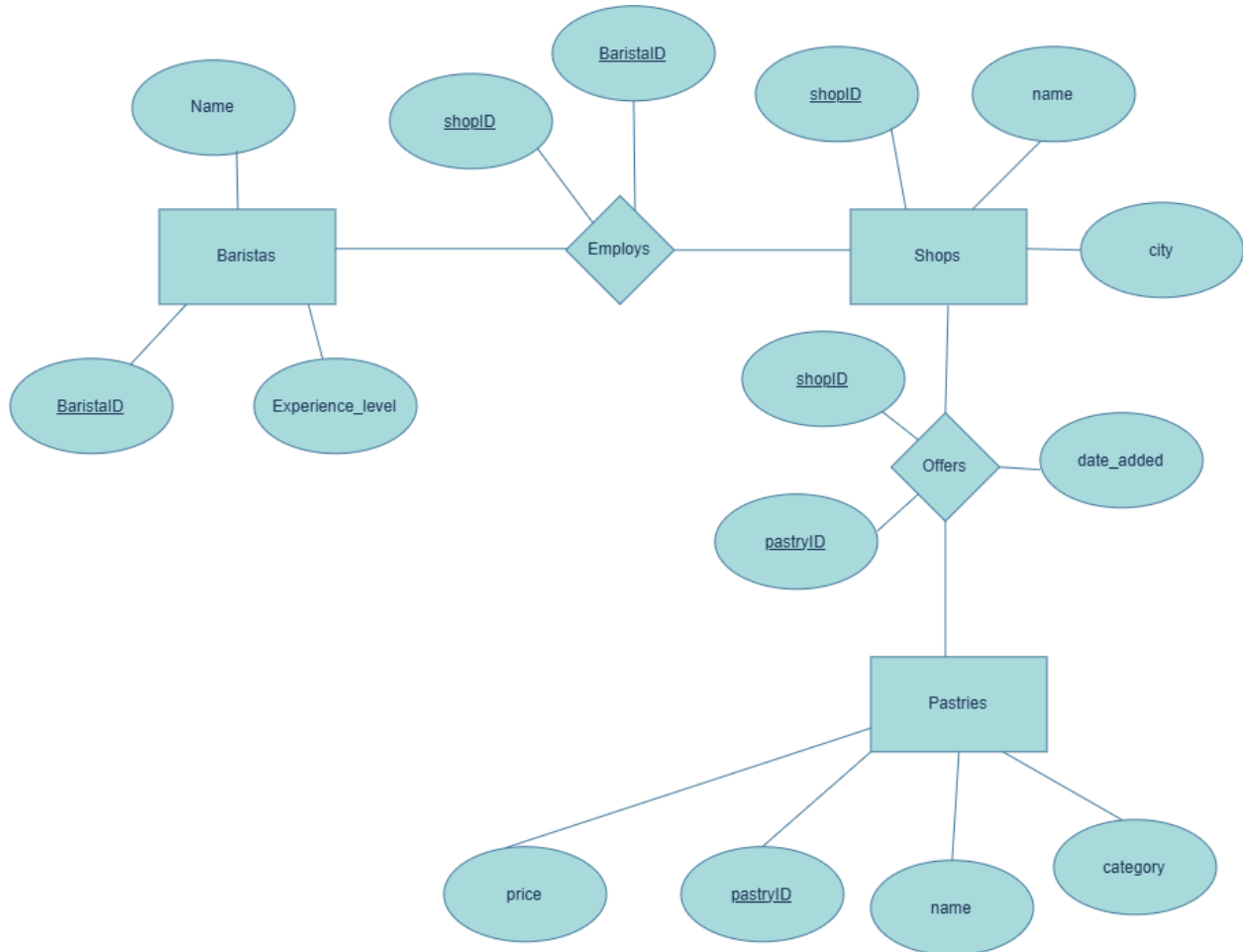


NAME: DURO-AINA SAMUEL
ASSIGNMENT 2

1. ENTITY RELATIONSHIP DIAGRAM

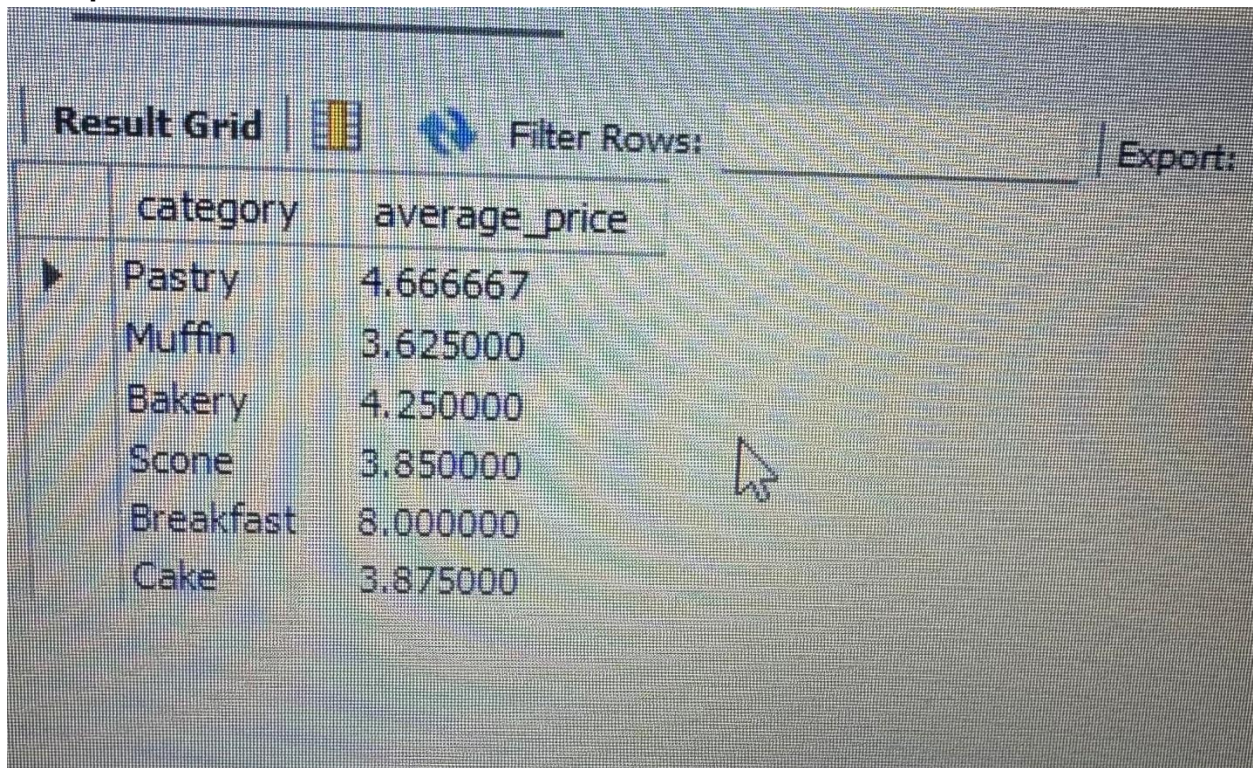


2. SQL

a) Average price of pastries

```
select category, avg(price) as average_price
from pastries
group by category;
```

Output



The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of a query, with columns 'category' and 'average_price'. The data is as follows:

	category	average_price
▶	Pastry	4.666667
	Muffin	3.625000
	Bakery	4.250000
	Scone	3.850000
	Breakfast	8.000000
	Cake	3.875000

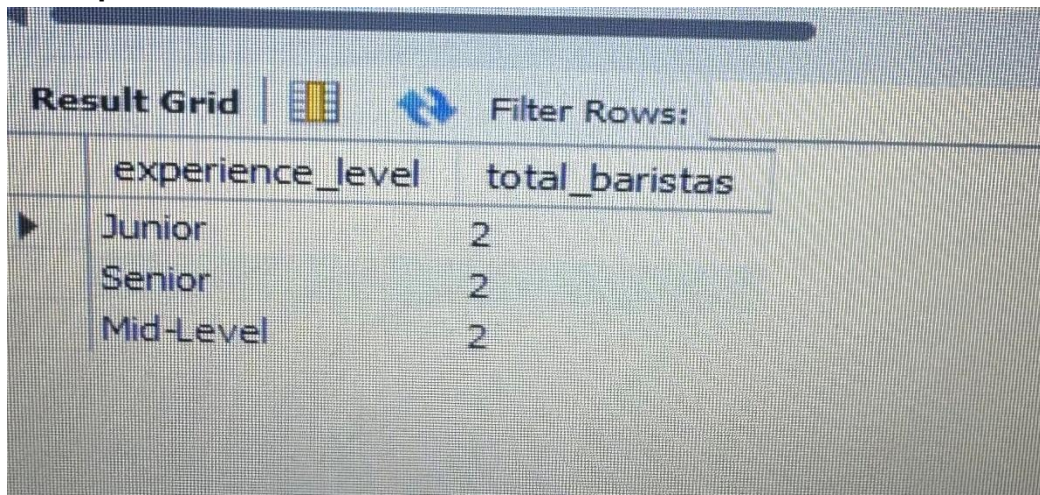
b) Total number of baristas

select experience_level, count(*) as total_baristas

from baristas

group by experience_level;

Output



The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of a query, with columns 'experience_level' and 'total_baristas'. The data is as follows:

	experience_level	total_baristas
▶	Junior	2
	Senior	2
	Mid-Level	2

c) Total number of shops in each city

```
select city, count(*) as total_shops
```

```
from shops
```

```
group by city;
```

Output



city	total_shops
Seattle	1
Portland	1
Austin	1
Chicago	1
Denver	1
Boston	1

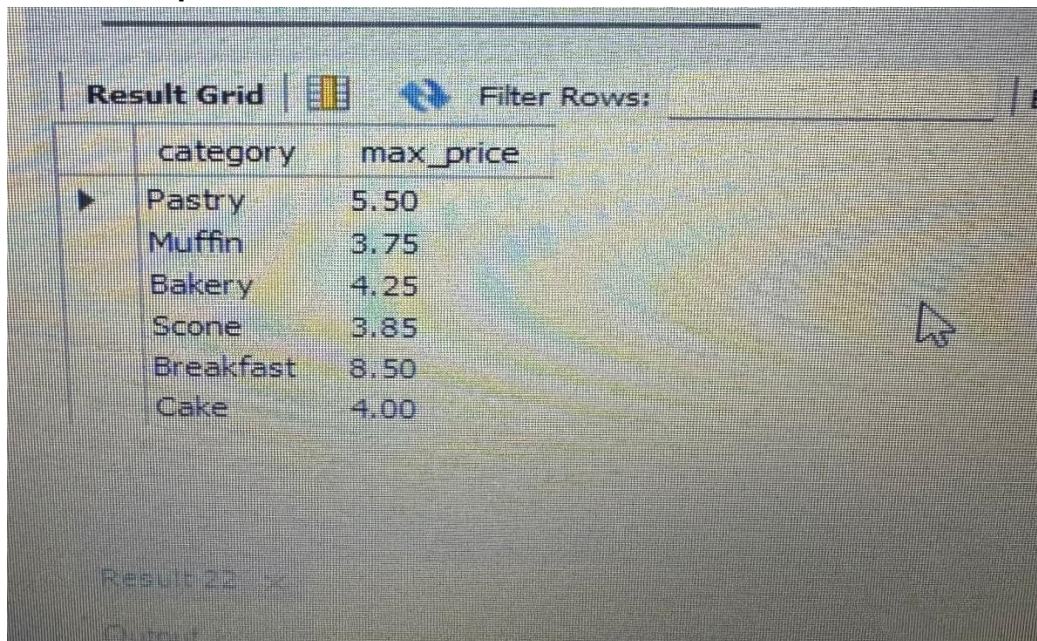
d) Maximum price among pastries

```
select category, max(price) as max_price
```

```
from pastries
```

```
group by category;
```

Output



The screenshot shows a database query result grid with the following data:

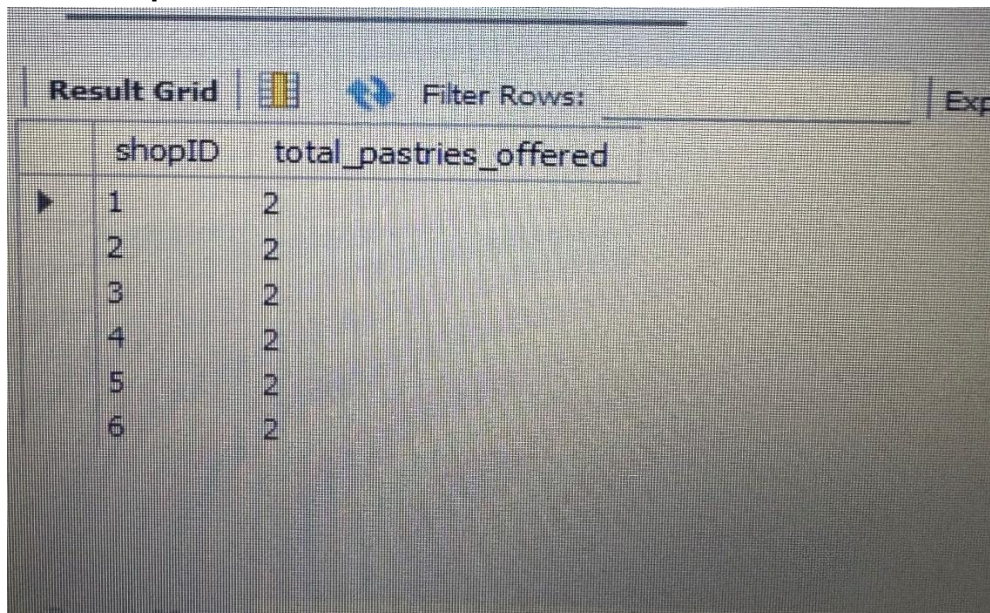
	category	max_price
▶	Pastry	5.50
	Muffin	3.75
	Bakery	4.25
	Scone	3.85
	Breakfast	8.50
	Cake	4.00

Below the table, the text "Result 22" and "Output" are visible.

e) Count of pastries added to shop

```
select shopID, count(pastryID) as total_pastries_offered  
from offers  
group by shopID;
```

Output



The screenshot shows a database query result grid with the following data:

	shopID	total_pastries_offered
▶	1	2
	2	2
	3	2
	4	2
	5	2
	6	2

f) **Name, category, and price of pastries whose price matches maximum price**

```
select name, category, price
```

```
from pastries p
```

```
where price = (
```

```
    select max(price)
```

```
    from pastries
```

```
    where category = p.category);
```

Output



The screenshot shows a database interface with a 'Result Grid' tab selected. It displays a table with three columns: 'name', 'category', and 'price'. The table contains seven rows of data. Above the table, there are icons for a grid, a refresh button, and a 'Filter Rows:' label. At the bottom left of the grid, it says 'pastries: 29'.

	name	category	price
▶	Almond Croissant	Pastry	5.50
	Blueberry Muffin	Muffin	3.75
	Cinnamon Roll	Bakery	4.25
	Chocolate Scone	Scone	3.85
	Blueberry Scone	Scone	3.85
	Breakfast Burrito	Breakfast	8.50
	Lemon Loaf	Cake	4.00

pastries: 29

g) **Unique shopIDs for offers that offered at least one pastry**

```
select distinct shopID
```

```
from offers
```

```
where pastryID in (
```

```
    select pastryID
```

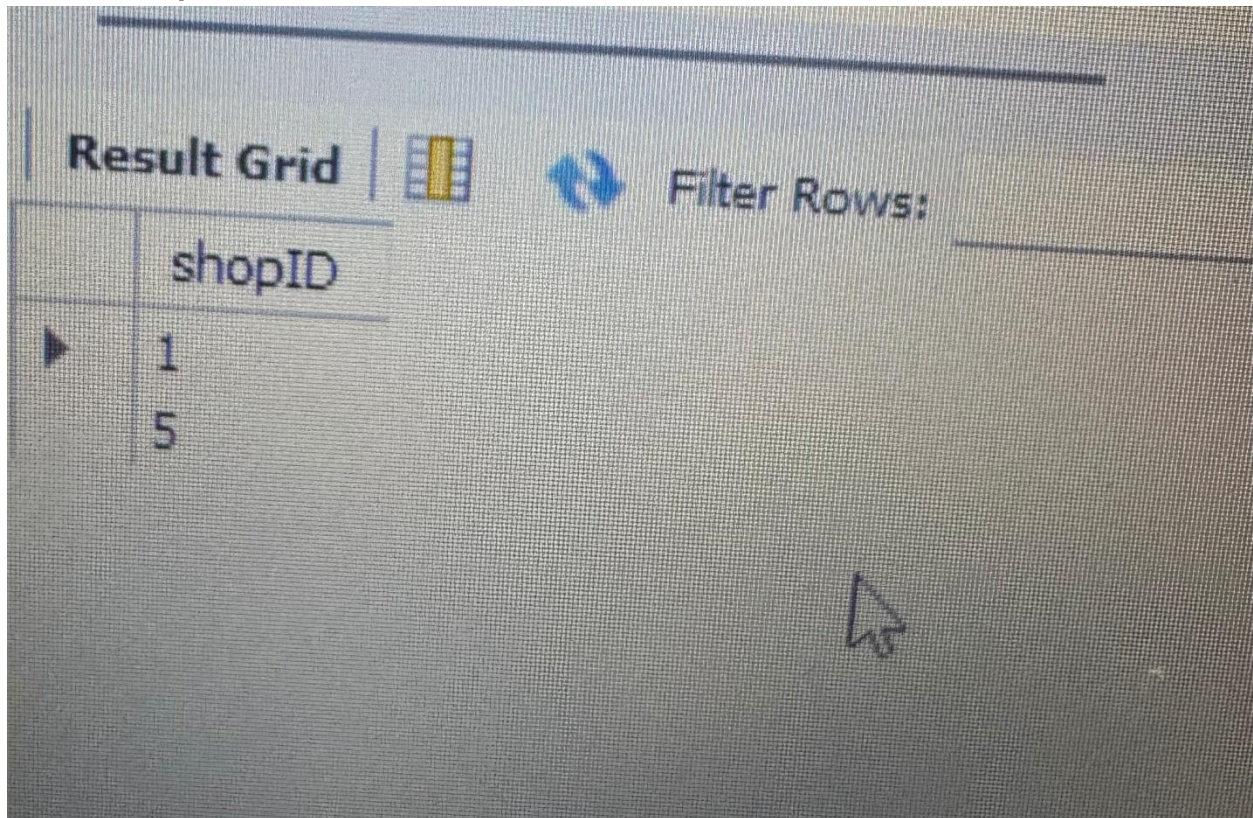


```

from pastries
where price > (
    select avg(price)
    from pastries
    )
);

```

Output



Result Grid	
	shopID
▶	1
	5

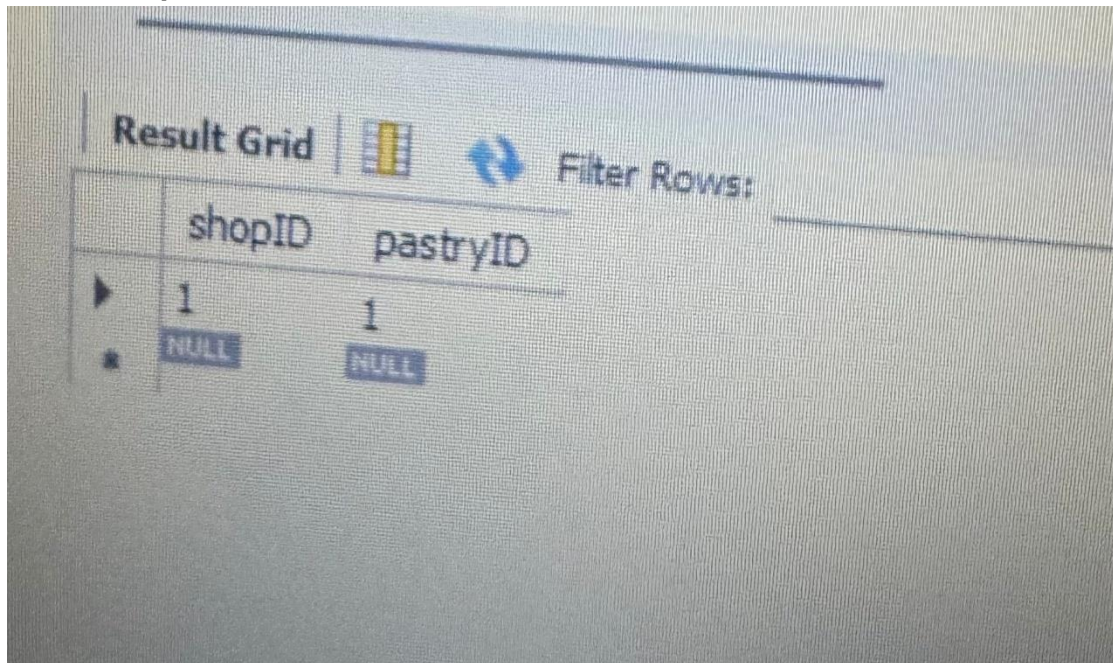
h) Shop ID and pastry ID for records in offers with the earliest date added

```

select shopID, pastryID
from offers
where date_added = (
    select min(date_added)
    from offers);

```

Output

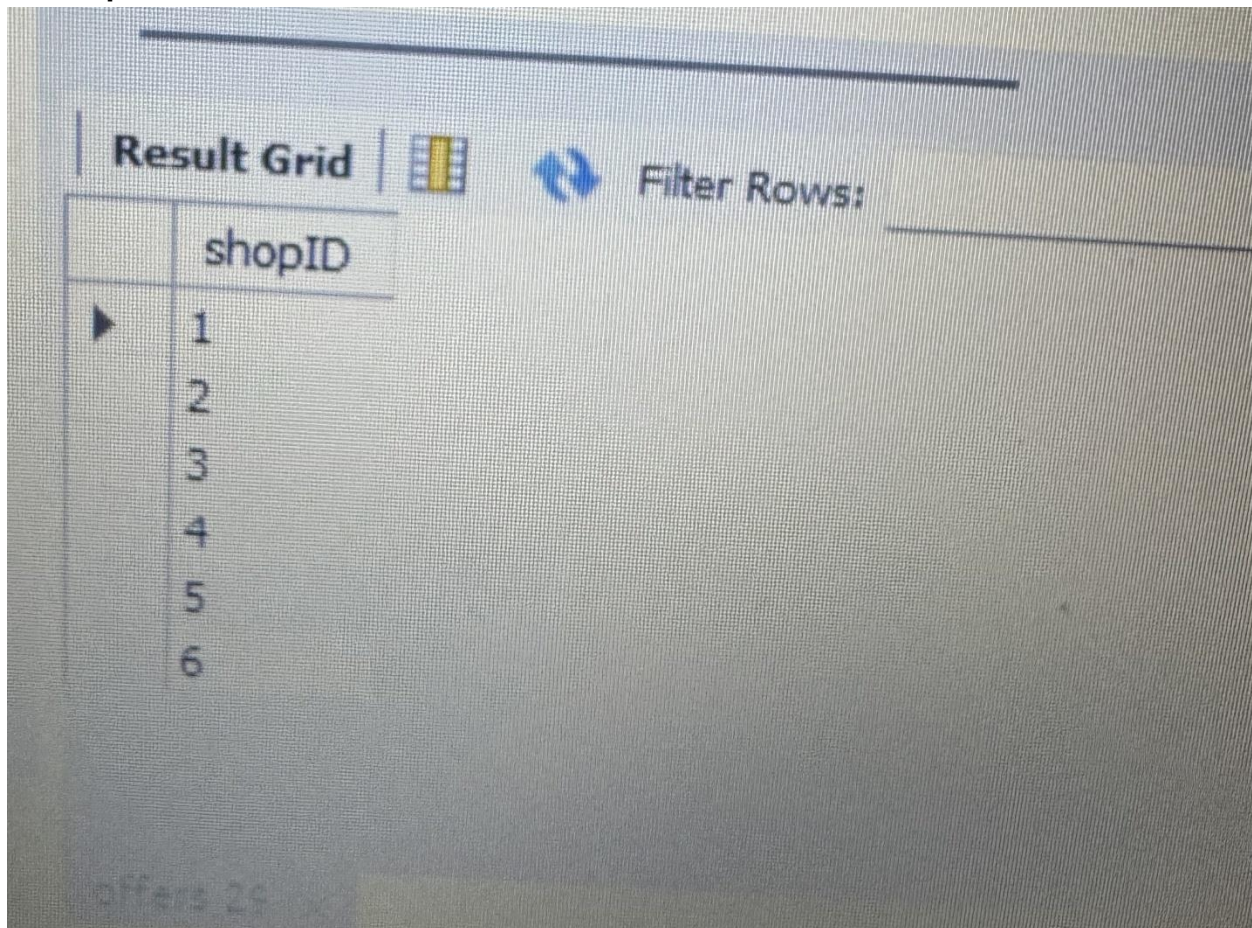


	shopID	pastryID
▶	1	1
★	NULL	NULL

i) Shop IDs offering the highest amount of pastries

```
select shopID
from offers
group by shopID
having count(*) >= all (
    select count(*)
    from offers
    group by shopID);
```


Output



The image shows a screenshot of a database application interface. At the top, there is a header bar with the text "Result Grid" and a small icon of a grid. To the right of the header bar, there is a blue circular arrow icon and the text "Filter Rows:". Below the header bar, there is a table with a single column labeled "shopID". The table contains six rows with the values 1, 2, 3, 4, 5, and 6. The first row is highlighted with a blue background. At the bottom left of the screenshot, there is a faint text "offers 29" and a small icon of a barista.

shopID
1
2
3
4
5
6

j) **Names of baristas who work at shops located in 'Seattle'**

select name

from baristas

where baristaID in (

select baristaID

from employs

where shopID in (

select shopID

from shops

where city = 'Seattle'

)

);

Output

Result Grid				Filter Rows:
	name			
▶	Emma Watson			
	Liam Chen			